

John Alfred Nathanael Chee

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INTERESTS

Understanding abstractions, applying them in practice. Writing simple and robust code.

Skills

Programming Languages – Haskell, Python, bash, SQL

Development Environment – vim, git, nix, Linux

PROFESSIONAL EXPERIENCE

Self-employed (Portland, Oregon)

Independent Software Specialist, Part-time, August 2023 – Present

- ▶ Developing an audiotheque app in the Protestant resourcement space, bringing Reformation authors and ideas into the 21st century.
- ▶ Also developing an offline-now, sync-later, privacy-first application to store all your memes.
- ▶ Provided ad hoc technical advising and support for Macintosh desktops and iPads, Google Workspace (Voice, Gmail, Groups primarily) for a 200+ user deployment.
- ▶ Contributed volunteer hours to the digital ministry of New Community Baptist Church, crafting and maintaining the website, YouTube live streams, and recordings.
- ▶ Maintainer of open-source Haskell projects: `haskell-cgi` (pure CGI bindings for Haskell) and `stats` (command-line statistics for plaintext numerical data) since 2016.
- ▶ Various GitHub contributions for proprietary and open-source projects: probabilistic data structures and automated build tools.

RealOptions (San José, California)

Optimal Health Coach, Part-time, February 2019 – September 2021

- ▶ Provided Optimal Health coaching to male patients at RealOptions Obria Medical Clinics by encouraging and advising STI and crisis pregnancy patients.
- ▶ Maintained a clean and organized clinic environment by sanitizing exam rooms, restocking supplies, and preparing client care packages with essentials like hygiene products and educational materials, ensuring a welcoming and efficient space for clients.
- ▶ Entered and updated client intake data, medical histories, and service records in EMR systems, adhering to HIPAA regulations, while troubleshooting minor system issues to support seamless clinic operations.
- ▶ Documented client information and medical notes in paper-based charts, organizing and archiving records to ensure compliance with confidentiality and retention policies.

Snowflake Computing (San Mateo, California)

Software Engineer Automation & Tools, April 2016 – August 2017

- ▶ Working with the Automation & Tools team at Snowflake, I initiated adoption of reproducible, versionable, reviewable Jenkins job descriptions with the `jobs dsl` plugin.
- ▶ Conducted the automated build and test of the main Snowflake products: database, user interface, and database drivers for both Windows and Unix-like operating systems.
- ▶ Secured & isolated Jenkins using its off-the-shelf LDAP security integration and an in-house Vault-inspired authentication tool.
- ▶ Managed & advised OSS teams transitioning from internal source & build tools to GitHub & Travis CI.
- ▶ Left Snowflake to pursue pre-medical coursework and apply to medical school.

Twitter (San Francisco, California)

Software Engineer, January 2013 – March 2016

- ▶ As a Software Engineer on the developer productivity team at Twitter I contributed to Pants Twitter's monolithic build system.
- ▶ As a member of the platform intelligence team I contributed to TSAR, an Apache Spark-like, Time Series Aggregator.
- ▶ During my time with Platform Intelligence I also took the opportunity to contribute to the Algebird project TSAR relied upon for its probabilistic data structures.
- ▶ On the chargeback team at Twitter I co-architected a data warehouse to store facts related to usage and user and project identification. Additionally, I built a simple data gathering system that extracted, transformed, and loaded raw chargeback-related data (CPU, RAM, telemetry) from a diverse population of systems to vertica and mysql data warehouses.

MyLife (Mountain View, California)

Software Engineer, May 2010 – January 2013

- ▶ As a Software Engineer, I accepted the task of development, automation and deployment of our search service to our front end team. This involved developing expertise with Hadoop, Lucene and an internal MapReduce flow control tool. I developed a scalable Hadoop-based geocoding solution for our document corpus. I advocated for the use of open standards (HTTP, JSON and Thrift) and modern source control (Git).
- ▶ As MyLife's current and future needs became apparent, I developed a low-latency, highly available and scalable search solution based on Cassandra and Solr. MyLife's new search platform removed the need for deployment of Lucene indexes, which effectively eliminated high-latency delivery of our search service. The search service was always up to date within the past 10 minutes, down from monthly previously. The new search service was an HTTP-based service-oriented-architecture written primarily in Python using capistrano for deployment and Supervisor for administration.
- ▶ Implemented a JSON-sensitive diff to immensely simplify data quality checking.
- ▶ Combined Haskell's QuickCheck library and siege to benchmark whole system performance with property-based test cases.
- ▶ I developed a scalable geocoding system using MapReduce and an in memory cache to vastly improve performance and minimize network calls.

Prospectus Group (Portland, Oregon)

Systems Administrator, March 2006 – August 2006

- ▶ As a systems administrator, I installed and configured a small secure network of 5 Linux servers. I also managed our production website with a Perl-based Content Management System.

Pacific Benefits Group

Information Technology Intern, November 2005 – March 2006

- ▶ As an information technology intern, I built, installed, and configured 50 workstations. Additionally, I managed specialized insurance data mining tools.
- ▶ Co-managed the company email server and backups.

RESEARCH

Portland State University, Department of Computer Science (Portland, Oregon)

Research Assistant: Project 10-9, January 2007 – August 2009

- ▶ As a research assistant on Project 10-9, an Open Voice Bridge, I worked on a team to integrate CMUSphinx with an existing MS Access query tool. 10-9 is primarily developed using Java and Subversion.
- ▶ Demonstrated Project 10-9 to focus groups and integrated feedback into future versions.

Portland State University, Department of Civil Engineering (Portland, Oregon)

Graduate Research Assistant: ITS Lab, January 2008 – August 2009

- ▶ As a graduate research assistant, I classified and discovered data gaps, automated daily discovery of erroneous readings, and began work on a time dependent view of overall transit system health. I primarily used PHP, SQL, and Haskell to discover and present the information.
- ▶ Co-authored and presented Transit Performance Measurement and Arterial Travel Time Estimation Using Archived AVL Data.

- Managed several database tables with millions of records.

Portland State University, Department of Computer Science (Portland, Oregon)

Computer Science Tutor, April 2006 – December 2006

- As a computer science tutor, I helped undergraduate students with primary courses including: compilers, data structures, and computational structures. Most questions were about representing data structures in C++ and functional programming in Standard ML.

Portland State University, Department of Civil Engineering (Portland, Oregon)

Undergraduate Research Assistant: ITS Lab, November 2006 – December 2008

- As an undergraduate research assistant on the Portland Oregon Regional Transportation Archive Listing (PORTAL) project, I created a web application using AJAX, PostgreSQL, PHP that displayed archived, geocoded, public transit system data on a Google map. I also helped process, aggregate, and query traffic and geospatial data using Python.

EDUCATION

San Mateo Community College District (Redwood City, California)

Student, August 2017 – May 2019

- Completed pre-medical classes at Cañada college and UC Berkeley.
- Biology courses: Principles of Biology, Biology of organisms, Cell & molecular biology, Biological chemistry (at UC Berkeley extension).
- Chemistry courses: General chemistry, Organic chemistry.
- Applied and interviewed for MD, Post-baccalaureate, and Master's programs at medical schools for the '19, '21, and '22 application cycles.

Portland State University, Portland, Oregon

M.S. in Computer Science; Advisor: Bart Massey, GPA: 3.78, June 2009

Portland State University, Portland, Oregon

B.S. in Computer Science; Minor: Mathematics, *Cum Laude*, March 2008

Portland Community College, Portland, Oregon

A.S., June 2004

PUBLISHED PROCEEDINGS

Berkow, M., Chee, J., Bertini, R.L., and Monsere, C., "Transit Performance Measurement and Arterial Travel Time Estimation Using Archived AVL Data," Compendium of Technical Papers, Institute of Transportation Engineers, District 6 Annual Meeting, Portland, Oregon, July 15 – 18, 2007.

PROFESSIONAL ACTIVITIES

Professional Memberships

Association for Computing Machinery, 2006 – 2008

Conferences & Gatherings Attended

BayHac, Bay Area Haskell Hackathon, 2011 – 2014
 Haskell Hackers at the Hacker Dojo, 2012 – 2014
 International Conference on Functional Programming, 2013
 Bay Area Haskell Users Group, 2011 – 2012
 Commercial Users of Functional Programming, 2010

AWARDS

President's List, Portland State University, Winter 2007 – Summer 2007
 Dean's List, Portland State University, Spring 2006 – Fall 2006

SERVICE

Professional Service

Secretary, Pre-med/Biomedical Research Club, Cañada College, January 2018 – May 2018
Treasurer, ACM Student Chapter, Portland State University, April 2006 – March 2008

Community Service

Patient Advocate, RealOptions Obria Medical Clinics, February 2019 – September 2021
Awana T&T Director, New Community Baptist Church, August 2017 – Present
Elementary Sunday School Teacher, New Community Baptist Church, May 2013 – Present
Classroom Assistant: ESL, Sunset Presbyterian Church, April 2004 – April 2006

Last updated: November 6, 2025

<https://github.com/cheecheeo/resume/blob/master/cv.pdf?raw=true>